

AI-Driven Support System for Neurologists: Integrating Neuro-Psychiatric Data for Early Detection of Psychiatric Disorders

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Introduction

- **Importance of Mental Health**

Key to resilience:- affects emotions, thinking, and life quality

WHO:- 1 out of 4 may experience mental health disorders, 14% of the global disease burden

- **Challenges in Mental Health Care**

Rising cases burden psychiatrists.

Complex diagnoses need behavioural, neurological, and social data

Lack of timely, personalized care causes delays

- **AI-Driven Solution**

Uses real-time data from behavioural and neurological sources.

Innovative integration of AI with neuroscience and psychiatry for proactive care

Reduces provider workload:- delivers precise, adaptive, and timely care.

Background of the study



- Completed an Independent Study in Level 3 on Detecting Brain Lesions to Identify Multiple Sclerosis.
- Experiencing Sudden Disability Due to a Neurological Disorder
- Seeking Psychiatric Help



Problem in Brief

- **Field Disconnection**

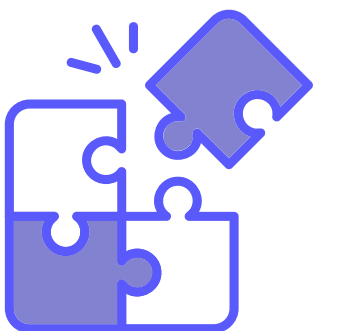
Knowledge from neurology and psychiatry isn't well-shared across disciplines, slowing down treatment innovation.

- **Need for Integrated Models**

The lack of comprehensive models that merge neurology and psychiatry insights limits our understanding of these conditions.

- **Limited Scope of Neurologists in Identifying Psychiatric Disorders**

Neurological assessments alone often fail to reveal signs of psychiatric conditions. Neurologists typically need to refer patients to psychiatrists for suspected mental health issues.



Aim of our Research

“The aim of this research is to develop an AI-driven support system that integrates neurological data with psychiatric evaluations to detect psychiatric disorders in neurologically affected patients.”



Research Objectives

- Explore a Suitable Dataset for Analyzing Neurological and Psychiatric Data.
- Explore Suitable AI and ML Techniques for Identifying Psychiatric Disorders.
- Study the mental and neurological disorders Relationship Between Neurological and Psychiatric Conditions.
- System Design and Development.
- Evaluate the developed system according to the evaluation criteria.



Proposed Solution - Modules



1. Dementia Detection Module (Wathsalya Gamage - 204058T)

Monitor cognitive changes that may indicate the early stages of dementia or neurological disorders.



2. Depression Detection Module (Oshani Kumarathunga - 204110X)

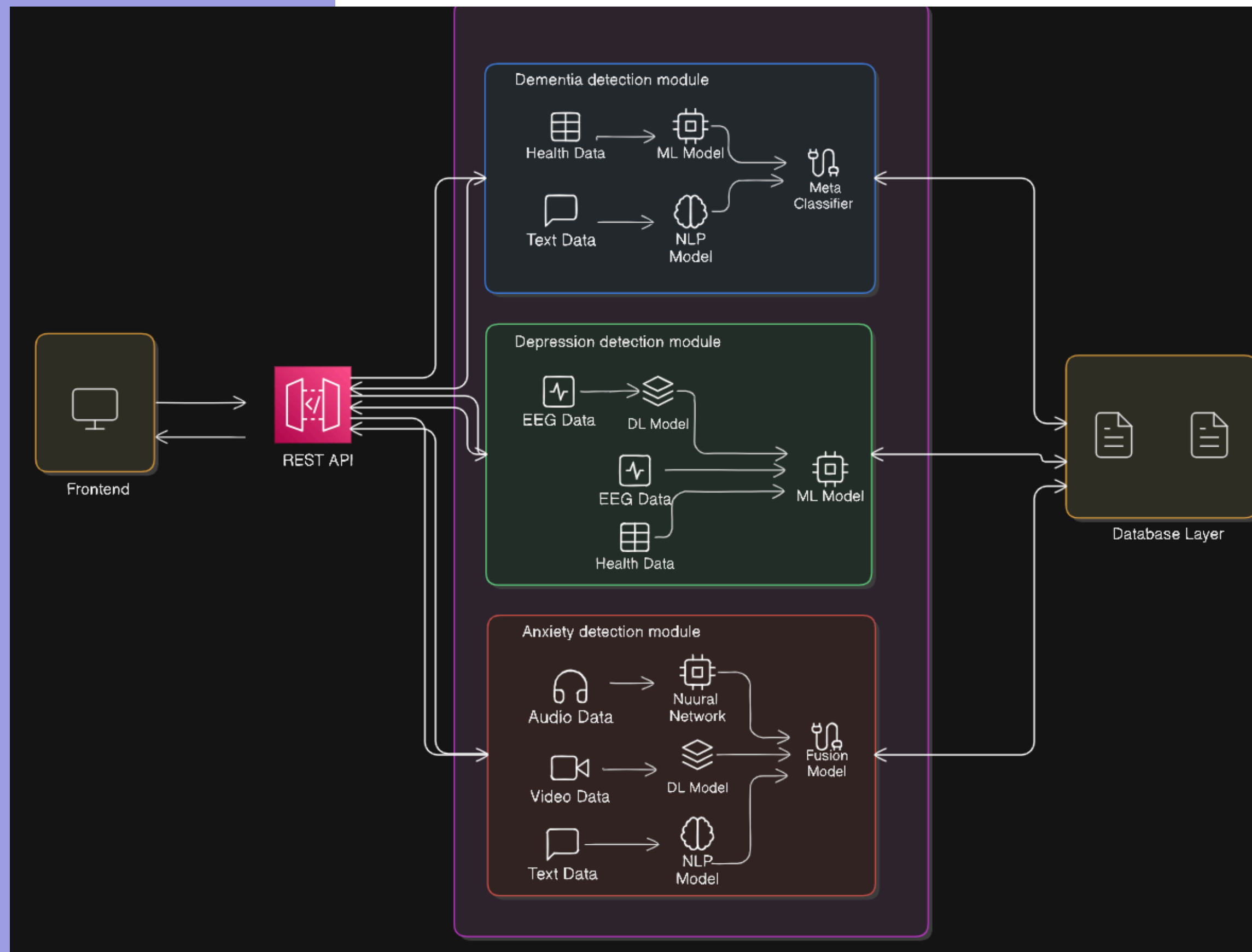
Identify depressive symptoms early, as depression frequently coexists with neurological disorders and affects the patient's quality of life and disease progression.



3. Anxiety Detection Module (Buddhima Kaushalya - 204111C)

Detect early signs of anxiety and type of anxiety, which is common in neurological disorders patients and can worsen overall symptoms.

High Level Architecture



Module 1 – Dementia Detection Module – 204058T

Research Gap: Integrating Text and Health Data for Dementia Detection

- Traditional dementia detection models rely on either linguistic patterns or medical data, but not both.
- Cognitive decline is reflected in both speech patterns and biomedical indicators.
- Current research lacks a multi-modal approach that combines text-based and health-based models.
- Our approach:
 - Train separate AI models for text and medical data.
 - Use ensemble learning to combine predictions for higher accuracy.
- Impact: More reliable, data-driven early dementia detection.

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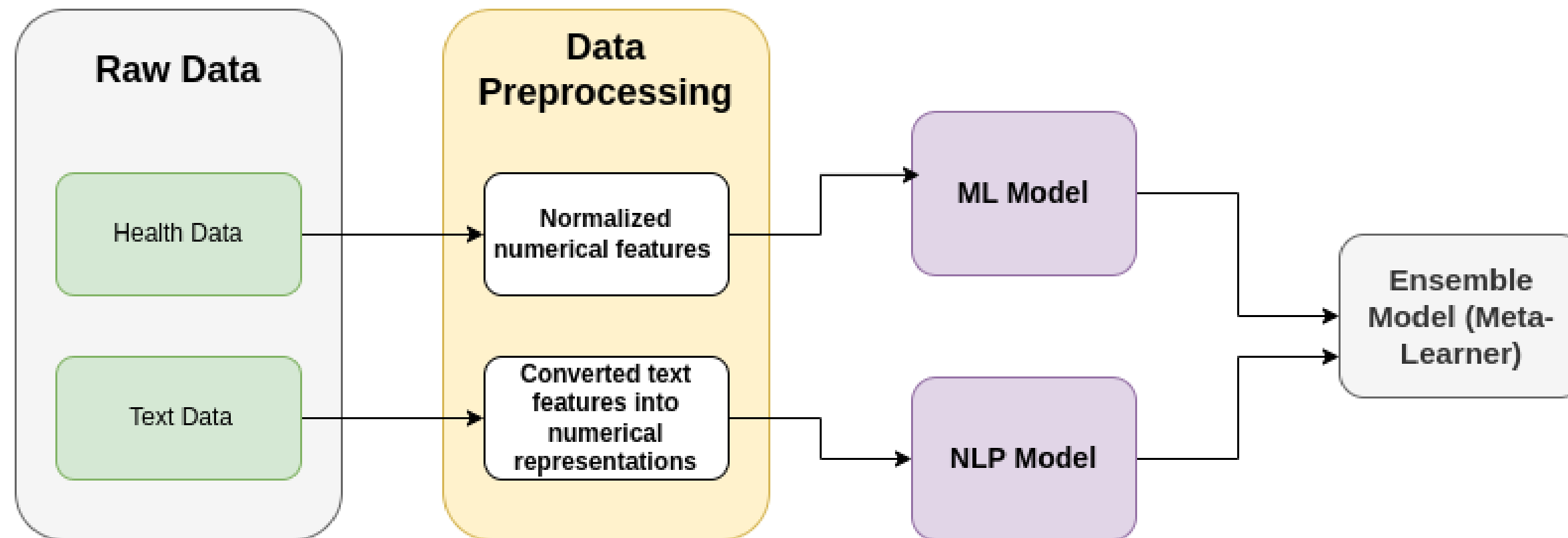
Initial Work and Final Direction

- Explored Diagnosis Methods:
 - **MRI-based**
 - Existing models: >95% accuracy
 - Requires high-quality imaging and domain expertise
 - **EEG-based**
 - Existing models: >90% accuracy
 - Limited access to reliable datasets
 - **Retina-based**
 - No publicly available datasets found
 - High potential but not feasible for this project
- Final Direction focuses on:
 - Text-based data
 - Structured clinical data

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System Workflow

- Step 1: Process text-based features using an NLP model.
- Step 2: Train a health-based classifier on medical features.
- Step 3: Combine predictions from both models using ensemble learning.
- Step 4: Final output – improved dementia detection accuracy.



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Implementation Approach

- Text-Based Model:
 - Fine-tune BERT on labeled dementia data for better linguistic pattern detection.
 - Extract final embeddings and pass them to a classifier for prediction.
- Health-Based Model:
 - Optimize hyperparameters for better prediction accuracy.
 - Perform feature selection to identify the most impactful health indicators.
- Ensemble Learning (Fusion Model):
 - Combine predictions from BERT and XGBoost.
 - Use a meta-classifier (e.g., another XGBoost or simple neural network).
 - Evaluate the ensemble model on test data for final dementia classification.

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Data Collection and Preprocessing

- **Dataset 1 (Text Data)**

- Removed special characters, stopwords, and standardized text format.
- Converted text features into numerical representations for model training.

- **Dataset 2 (Health Data)**

- Handled missing values through imputation techniques.
- Normalized numerical features for consistent scaling in ML models.

- **Data Validation & Cleaning:**

- Addressed inconsistencies, removed outliers, and ensured data quality.
- Checked for imbalanced classes in both datasets.

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Clinical Data – Binary Classification

- **Models Tested:**

- XGBoost → Accuracy: 92%
- Random Forest → Accuracy: 95%

- **Details:**

- Task: Binary classification (Dementia Positive vs Negative)
- Result: Random Forest performed best with 95% accuracy
- Data: Medical/clinical records

```
XGBoost Classifier:  
Cross-Validation Scores: [0.89344262 0.8888  
Mean Cross-Validation Score: 0.908729676853  
Accuracy: 0.9162790697674419
```

```
Random Forest Classifier:  
Cross-Validation Scores: [0.92213115 0.9  
Mean Cross-Validation Score: 0.929282196  
Accuracy: 0.9511627906976744
```

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Clinical Data – Severity Classification

- **Steps:**
 - Used K-Means Clustering to group dementia-positive individuals
 - Created 3 severity levels (Mild, Moderate, Severe)
 - Total classes: 4 (including non-dementia class)
- **Multiclass Classification:**
 - Tested various models on new 4-class dataset
 - Best Accuracy: 84.186%

Model Evaluation Results:						
model	accuracy	precision	recall	f1	roc_auc	
XGBClassifier	84.186	84.2925	84.186	84.1973	95.7117	
Gradient Boosting Classifier	80	82.1841	80	80.4022	95.1979	
Random Forest	79.3023	78.9324	79.3023	78.9819	92.4612	
Decision Tree	72.5581	73.9171	72.5581	73.0424	78.3076	
Support Vector Machine	37.6744	56.2965	37.6744	37.1506	nan	
K-Nearest Neighbors	27.907	46.8194	27.907	31.0315	54.2104	

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Clinical Data – Fine tuning the model

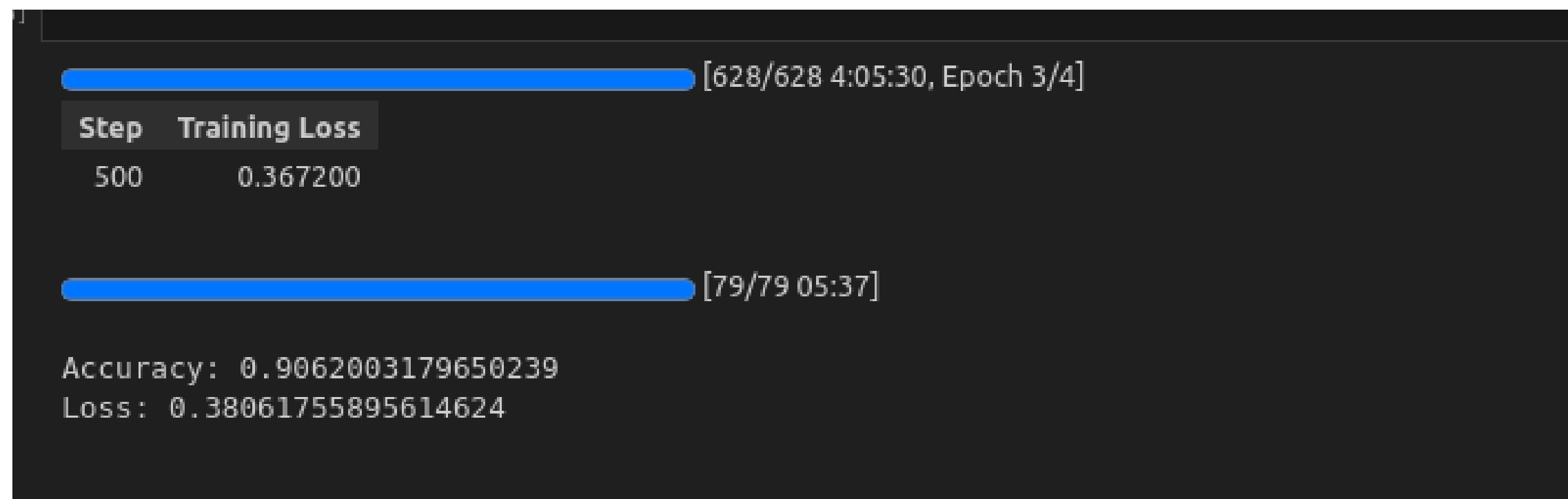
- Hyperparameter tuning and evaluation of the XGBoost classifier.
- Used RandomizedSearchCV and GridSearchCV to optimize the model's performance.
- Incorporated SMOTE to handle class imbalance.
- The best model is evaluated on the test set using metrics like accuracy, precision, recall, and F1-score.
- Detailed probability analysis and optimized pipeline are saved for future use.

Classification Report:				
	precision	recall	f1-score	support
0	0.93	0.94	0.94	254
1	0.91	0.88	0.89	67
2	0.89	0.93	0.91	45
3	0.95	0.91	0.93	64
accuracy			0.93	430
macro avg	0.92	0.92	0.92	430
weighted avg	0.93	0.93	0.93	430

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Text Data – NLP Model

- **Model Used:**
 - BERT (BertForSequenceClassification)
 - Task: Binary classification (Dementia-related text vs Normal)
 - Accuracy: 90%
- **Approach:**
 - Preprocessed text data
 - Used BERT model for classification



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Text Data – Feature Extraction

- Extracted linguistic and cognitive features from the dataset.
- Processed text data using spaCy and calculated features like Type-Token Ratio (TTR), Brunet's Index, average word length, part-of-speech ratios, sentence complexity, and idea density.
- Additional features include counts of key elements, irrelevant details, pauses, and repair rates in the text.
- The processed dataset is saved for further analysis.

```
apply features to full dataset
"TTR" = df["combined_text"].apply(calculate_ttr)
"Brunet_Index" = df["combined_text"].apply(calculate_brunet)
"Avg_Word_Length" = df["combined_text"].apply(calculate_avg_word_length)

_pos_ratios = df["combined_text"].apply(calculate_pos_ratios)
"NOUN_ratio" = _pos_ratios.apply(lambda x: x["NOUN_ratio"])
"VERB_ratio" = _pos_ratios.apply(lambda x: x["VERB_ratio"])
"PRONOUN_ratio" = _pos_ratios.apply(lambda x: x["PRONOUN_ratio"])

_complexity = df["combined_text"].apply(calculate_sentence_complexity)
"Subordinate_Clauses" = _complexity.apply(lambda x: x["subordinate_clauses"] if x else 0)
"Parse_Tree_Depth" = _complexity.apply(lambda x: x["parse_tree_depth"] if x else 0)

"Idea_Density" = df["combined_text"].apply(calculate_idea_density)

_key_elements = ["boy stealing cookies", "sink overflowing", "mother", "kitchen", "cookies"]
_irrelevant_details = ["dog", "cat", "car", "tree"]

"Key_Elements_Described" = df["combined_text"].apply(lambda x: count_key_elements(x, key_elements))
"Irrelevant_Details" = df["combined_text"].apply(lambda x: count_irrelevant_details(x, irrelevant_details))
"Pauses" = df["combined_text"].apply(count_pauses)
"Repair_Rate" = df["combined_text"].apply(calculate_repair_rate)
```

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Text Data – Feature Extraction

- Created severity levels for dementia patients based on linguistic and cognitive features.
- It calculates composite scores like Disfluency_Ratio, Severity_Score, and Cognitive_Expression_Score from the dataset.
- KMeans clustering is applied to group dementia patients into three severity levels, which are then mapped to labels (1, 2, 3).
- The final dataset, including control group patients with severity 0, is split into training and testing datasets and saved.

Cluster means (for interpretation):

Severity_Cluster	Cognitive_Expression_Score	Disfluency_Ratio	Severity_Score
0	2.388518	0.368421	0.085169
1	2.289520	8.570000	1.981019
2	6.600171	0.634680	0.196063

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Text Data – Model Evaluation

- BERT-based multi-class classification model is evaluated for predicting dementia severity levels (0, 1, 2, 3) using text data.
- The BERT tokenizer is used to encode the text, and the model is fine-tuned using the Trainer API with specified training arguments.
- Its performance is evaluated using metrics like accuracy, classification report, and confusion matrix. The results provide insights into the model's effectiveness in classifying dementia severity levels.

```
print("Evaluation Results:")
print(results["eval_classification_report"])
print("\nConfusion Matrix:")
print(results["eval_confusion_matrix"])
print(f"\nAccuracy: {results['eval_accuracy']}")
```

Python

```
Evaluation Results:
{'0': {'precision': 0.9193548387096774, 'recall': 0.7916666666666666, 'f1-score': 0.8507462686567164, 'sup

Confusion Matrix:
[[57, 3, 12, 0], [1, 50, 1, 4], [4, 1, 13, 0], [0, 2, 1, 6]]

Accuracy: 0.8129032258064516
```

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Text Data – Model Evaluation

- The machine learning pipeline is fine-tuned to predict dementia severity levels using both linguistic features and BERT embeddings.
- BERT embeddings are extracted for three transcript types. These embeddings are combined with linguistic features, and hyperparameter tuning is performed using RandomizedSearchCV to optimize an XGBoost classifier.
- The model is evaluated using metrics like accuracy, precision, recall, and F1-score, with detailed probability analysis.

Classification Report:				
	precision	recall	f1-score	support
0	0.85	0.89	0.87	92
1	0.93	0.99	0.96	67
2	0.44	0.24	0.31	17
3	0.78	0.70	0.74	10
accuracy			0.85	186
macro avg	0.75	0.70	0.72	186
weighted avg	0.84	0.85	0.84	186
Confusion Matrix:				
[[82 4 4 2]				
[0 66 1 0]				
[13 0 4 0]				
[2 1 0 7]]				

Module 1 - Dementia Detection Module - 204058T

Meta Classifier - Tuning and Evaluation

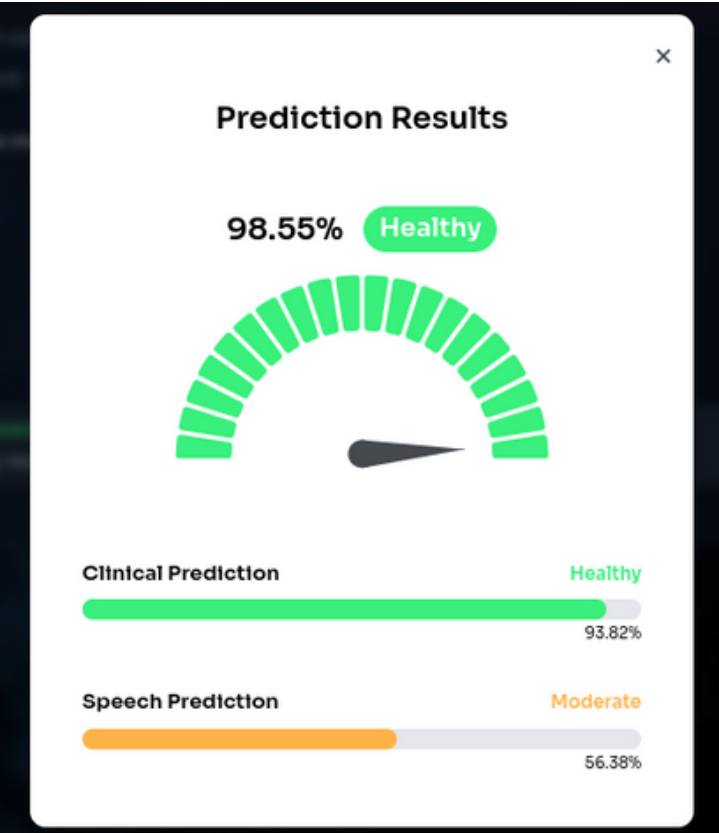
- Developed a meta-classifier to predict dementia severity by combining predictions from clinical and speech-based models.
- It enhances features with confidence scores, agreement metrics, and entropy, and handles class imbalance using SMOTE.
- The meta-classifier, trained using XGBoost or Random Forest, is fine-tuned with grid search and optionally calibrated for better probability estimates.
- The model is evaluated using metrics like accuracy, confusion matrix, and prediction confidence, and the final meta-classifier is saved

```
[54] predictions = predict_final(meta_model=train_results['meta_model'])

... Overall accuracy: 0.9577
      precision    recall  f1-score   support

    0       0.96       0.97       0.97        72
    1       0.95       0.98       0.97        43
    2       0.94       0.89       0.91        18
    3       1.00       0.89       0.94         9

   accuracy                0.96        142
  macro avg       0.96       0.93       0.95        142
 weighted avg       0.96       0.96       0.96        142
```



Module 1 - Dementia Detection Module - 204058T

Related Works

- [1] Shi, J., Gan, Q., Hanchrow, E., Bowles, A., Stanley, J., Bress, A. P., Cohen, J. B., & Alba, P. R. (2025). Accelerating clinical NLP at scale with a hybrid framework with reduced GPU demands: a case study in dementia identification. arXiv. Wu, C.-T., Huang, H.-C., Huang, S., Chen, I.-M., Liao, S.-C., Chen, C.-K., Lin, C., Lee, S.-H., Chen, M.-H., Tsai, C.-F., et al. (2021). Resting-State EEG Signal for Major Depressive Disorder Detection: A Systematic Validation on a Large and Diverse Dataset. Biosensors, 11(12), 499. <https://doi.org/10.3390/bios11120499>
- [2] Chen, Z., Zhang, H., Yang, X., Wu, S., He, X., Xu, J., et al. (2024). Extracting critical information from unstructured clinicians' notes data to identify dementia severity using a rule-based approach: feasibility study. JMIR Aging. Bader, M., Abdelwanis, M., Maalouf, M., & Jelinek, H. F. (2024). Detecting depression severity using weighted random forest and oxidative stress biomarkers. Scientific Reports, 14, Article 16328. <https://doi.org/10.1038/s41598-024-67251-y>
- [3] UConn Research Team. (2025). Real-world insights into dementia diagnosis trajectory and clinical practice patterns unveiled by natural language processing: development and usability study. JMIR Aging. Khan, S., Umar Saeed, S. M., Frnda, J., Arsalan, A., Amin, R., Gantassi, R., & Noorani, S. H. (2024). A machine learning based depression screening framework using temporal domain features of the electroencephalography signals. PLOS ONE, 19(3), e0299127. <https://doi.org/10.1371/journal.pone.0299127>

Module 2 - Depression Detection Module - 204110X

Introduction

Depression in Neurology Patients

- Neurological disorders such as Parkinson's disease, stroke, and epilepsy often lead to depression.
- Traditional tools like PHQ-9 and HAM-D may overlook subtle cognitive-affective symptoms in these patients.
- EEG offers a non-invasive, objective way to detect depression-related brain activity.

Why AI for Depression Detection in Neurology?

- Early Detection – Identifies hidden patterns in EEG for early diagnosis.
- Efficiency – Enables fast, automated, and scalable screening.
- Accuracy – Combines EEG with clinical data (e.g., PHQ-9) for better results.

Module Goal

- Fine-tune EEG foundation models (e.g., CBraMod) for accurate depression classification using resting-state EEG.
- Integrate band power features + PHQ-9 scores to support severity grading.
- Deliver a decision-support system for neurologists based on data-driven mental health insights.



Module 2 - Depression Detection Module - 204110X

Research Gap Addressed

1. EEG-Based Depression Detection with Deep Learning

Traditional diagnosis is subjective.

We use EEG signals and deep learning (CBraMod) for objective detection.

2. Fine-Tuned Foundation Model (CBraMod)

We leverage a pre-trained EEG model and fine-tune it to recognize depression-specific brain patterns.

3. Severity Level Classification

Most studies stop at binary detection.

We go further by predicting depression severity using EEG band power features.

Module 2 - Depression Detection Module - 204110X

Foundation Models for EEG

Large deep learning models are pretrained on diverse EEG datasets to learn general neural patterns.
They support:

- Reduce the need for large task-specific datasets
- Improve performance on downstream tasks
- Enable transfer learning in neuroscience and clinical applications

Some examples for EEG Foundation Models

- CBraMod (Jiquan Wang al., 2025)
- LaBraM (Jiang et al., 2024)
- BIOT (Yang et al., 2023) i
- ST-Transformer (Song et al., 2021)
- FFCL(Li et al., 2022)

CBraMod – Cross-Brain Foundation Model

Feature	Description
• Criss-Cross Transformer	Separates spatial (electrodes) & temporal (signal) modeling via dual attention
• Asymmetric Positional Encoding	Adapts to varying EEG channel layouts & lengths
• Masked EEG Pretraining	Learns by reconstructing masked EEG patches (like BERT)

Module 2 - Depression Detection Module - 204110X

Data Sets

- **MODMA**: 128-channel resting-state EEG with PHQ-9 scores and clinical labels
- **Mumtazz**: 64-channel EEG with binary labels (MDD vs. HC)
- **BMC**: 64-channel EEG from adolescents, labeled by PHQ-9 into minimal, mild, and moderate depression groups
- Data Format: Resting-state EEG, segmented into 5-sec windows
- Label Types:
 - Binary diagnosis (MDD/HC – Mumtazz)
 - PHQ-9 severity levels (mild/moderate/severe – MODMA, BMC)

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Modeling Progression and Final Pipeline

Step	Approach	Result	Note
1	MODMA Binary Classification Using CBraMod features	Accuracy ~65% Balanced Accuracy: 0.6458	Poor generalization
2	MODMA Clustering (PHQ-9)	Clusters too close Mean PHQ: 18.3 vs 18.8	PHQ severity not captured
3	MODMA Multi-Class Severity	Accuracy: 19% Macro F1: 0.12	Inconsistent feature-to-label mapping
4	MODMA PHQ Regression (LightGBM)	MAE: 3.35 $R^2 = 0.58$	Not reliable for scoring
5	MUMTAZZ Binary Classification CBraModMumtaz (Fine-Tuned)	Accuracy: 86.6% F1: 0.8487	Final Pipeline
6	Severity Classification Band Power + Randome Forest	Patient Accuracy: 84.21%	Final pipeline

Module 2 - Depression Detection Module - 204110X

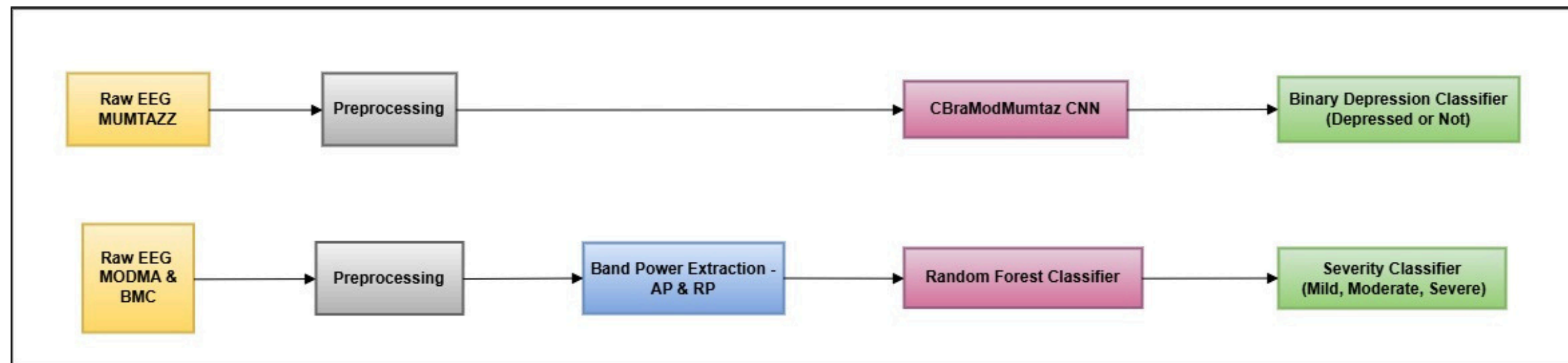
Final Implementation Approach

EEG-Based Model:

- Fine-tune CBraMod on labelled MUMTAZZ EEG data to capture depression-specific brain patterns.
- Train the model end-to-end with a built-in classifier to predict binary depression labels.
- Use spatiotemporal EEG inputs reshaped as $[19 \times 5 \times 200]$ segments.

Band Power-Based Model:

- Extract frequency-domain features (Delta, Theta, Alpha, Beta) from each 5-second EEG segment.
- Perform feature normalisation and selection to retain the most informative bands.
- Train a Random Forest classifier to predict severity levels.



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EEG Preprocessing Method

Step	Details
• Channel Selection	19 standard EEG channels (e.g., Fp1, F3, Cz, O1)
• Resampling	Downsampled to 200 Hz
• Filtering	Bandpass: 0.3–75 Hz Notch: 50 Hz (remove powerline noise)
• Segmentation	Split into non-overlapping 5s windows (1000 samples)
• Tensor Reshaping	Final shape: [segments, 19, 5, 200]
• Normalization	Signals converted to μV and divided by 100

```
selected_channels = [  
    'EEG Fp1-LE', 'EEG Fp2-LE', 'EEG F3-LE', 'EEG F4-LE', 'EEG C3-LE', 'EEG C4-LE', 'EEG P3-LE',  
    'EEG P4-LE', 'EEG O1-LE', 'EEG O2-LE', 'EEG F7-LE', 'EEG F8-LE', 'EEG T3-LE', 'EEG T4-LE',  
    'EEG T5-LE', 'EEG T6-LE', 'EEG Fz-LE', 'EEG Cz-LE', 'EEG Pz-LE'  
]
```

```
for split in ['train', 'val', 'test']:  
    for file in files_dict[split]:  
        raw = mne.io.read_raw_edf(os.path.join(root_dir, file), preload=True)  
        # Pick channels and reorder  
        raw.pick_channels(selected_channels, ordered=True)  
        raw.resample(200) # resample to 200Hz  
        raw.filter(l_freq=0.3, h_freq=75) # bandpass filter  
        raw.notch_filter(freqs=50)  
  
        eeg_df = raw.to_data_frame().iloc[:, 1:].values # remove time column  
  
        n_points, n_ch = eeg_df.shape  
        window_size = 5 * 200 # 5 seconds * 200 Hz = 1000 points  
  
        # Trim remainder for perfect segment division  
        remainder = n_points % window_size  
        if remainder != 0:  
            eeg_df = eeg_df[:-remainder]  
  
        segments = eeg_df.reshape(-1, window_size, n_ch)  
        segments = segments.transpose(0, 2, 1) # (num_segments, channels, time)  
  
        label = 1 if 'MDD' in file else 0  
  
        with db.begin(write=True) as txn:  
            for i, segment in enumerate(segments):  
                key = f"{file[:-4]}_{i}".encode()  
                data_dict = {'sample': segment.astype(np.float32), 'label': label}  
                txn.put(key, pickle.dumps(data_dict))  
            dataset_keys[split].append(key.decode())
```

Module 2 - Depression Detection Module - 204110X

Feature Extraction Methods

CBraMod (Fine-Tuned on Mumtazz data).

- Deep CNN-based model trained on 19-channel EEG inputs.
- Learns spatiotemporal EEG features from $[19 \times 5 \times 200]$ tensors.
- During fine-tuning on MUMTAZZ, CBraMod is adapted to better represent depression-specific patterns.
- Features are extracted internally through convolutional layers during training/inference.

Band Power Features

- Frequency-domain features (Delta, Theta, Alpha, Beta)
- Extracted from each 5-second segment
- Used for PHQ severity classification

```
class CBraModMumtaz(nn.Module):
    def __init__(self, param):
        super().__init__()
        self.backbone = CBraMod(d_model=200, n_layer=12, nhead=8) # adjust params if desired
        if param.use_pretrained_weights:
            self.backbone.load_state_dict(torch.load(param.pretrained_ckpt_path, map_location=param.device))

        self.backbone.proj_out = nn.Identity() # disable original head

        # Choose classifier architecture based on param
        if param.classifier == 'all_patch_reps_twolayer':
            self.classifier = nn.Sequential(
                Rearrange('b c s p -> b (c s p)'),
                nn.Linear(19 * 5 * 200, 200),
                nn.ELU(),
                nn.Dropout(param.dropout),
                nn.Linear(200, 1)
            )
        else:
            raise ValueError("Unsupported classifier type")

    def forward(self, x):
        features = self.backbone(x)
        out = self.classifier(features).squeeze(1)
        return out
```

```
def extract_band_powers(eeg, sfreq):
    """
    Calculate absolute and relative power for defined bands in one channel.

    eeg : 1D array (time points)
    sfreq : sampling frequency (e.g. 250Hz)
    """
    freqs, psd = welch(eeg, sfreq, nperseg=sfreq*2)
    total_power = np.trapz(psd, freqs)
    features = {}
    for band, (fmin, fmax) in BANDS.items():
        idx = np.logical_and(freqs >= fmin, freqs < fmax)
        ap = np.trapz(psd[idx], freqs[idx])
        rp = ap / total_power if total_power > 0 else np.nan
        features[f"{band}_ap"] = ap
        features[f"{band}_rp"] = rp
    features["total_ap"] = total_power
    return features
```

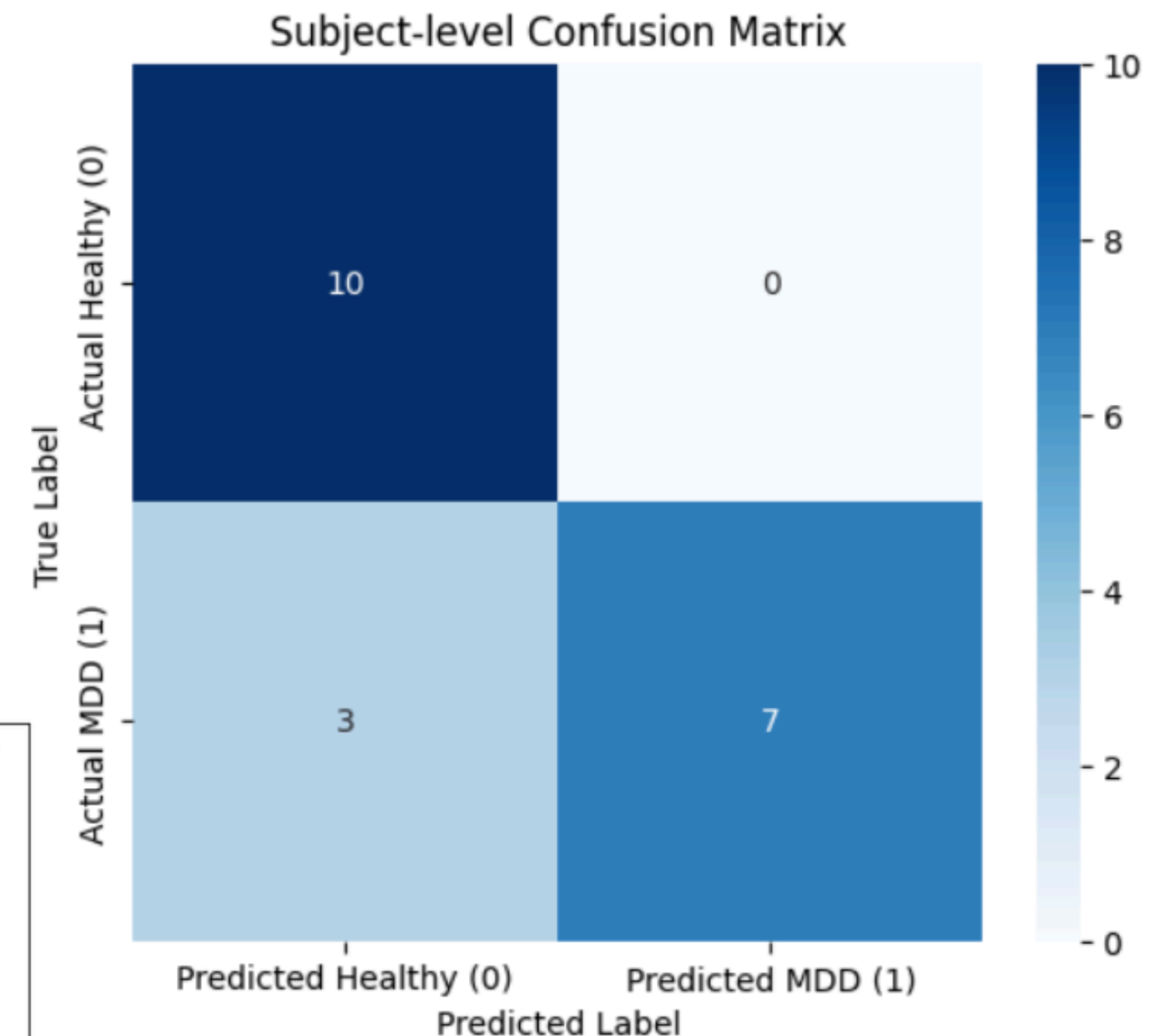
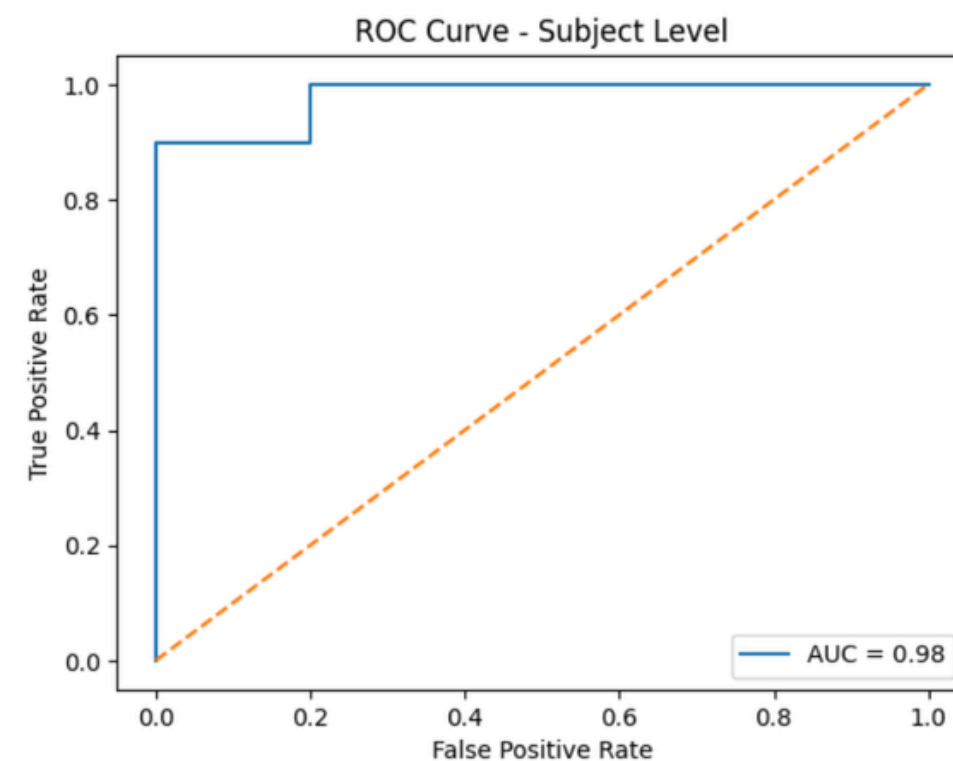
Module 2 - Depression Detection Module - 204110X

Mumtazz Binary Classification

- **Input:** EEG segments [19×5×200] after preprocessing
- **Model:** Fine-tuned CBraMod CNN on MUMTAZZ
- **Training strategy:** Early stopping, threshold tuning
- **Evaluation:**
 - Test Accuracy: 86.6%
 - F1 Score: 0.8487
 - Patient-level Accuracy: 85%

Test Accuracy: 0.8657
Precision: 0.9560, Recall: 0.7631, F1: 0.8487

Subject-level Accuracy: 0.8500
Precision: 1.0000, Recall: 0.7000, F1: 0.8235



Module 2 - Depression Detection Module - 204110X

EEG Severity Classification: Band Power + Random Forest

Input Features:

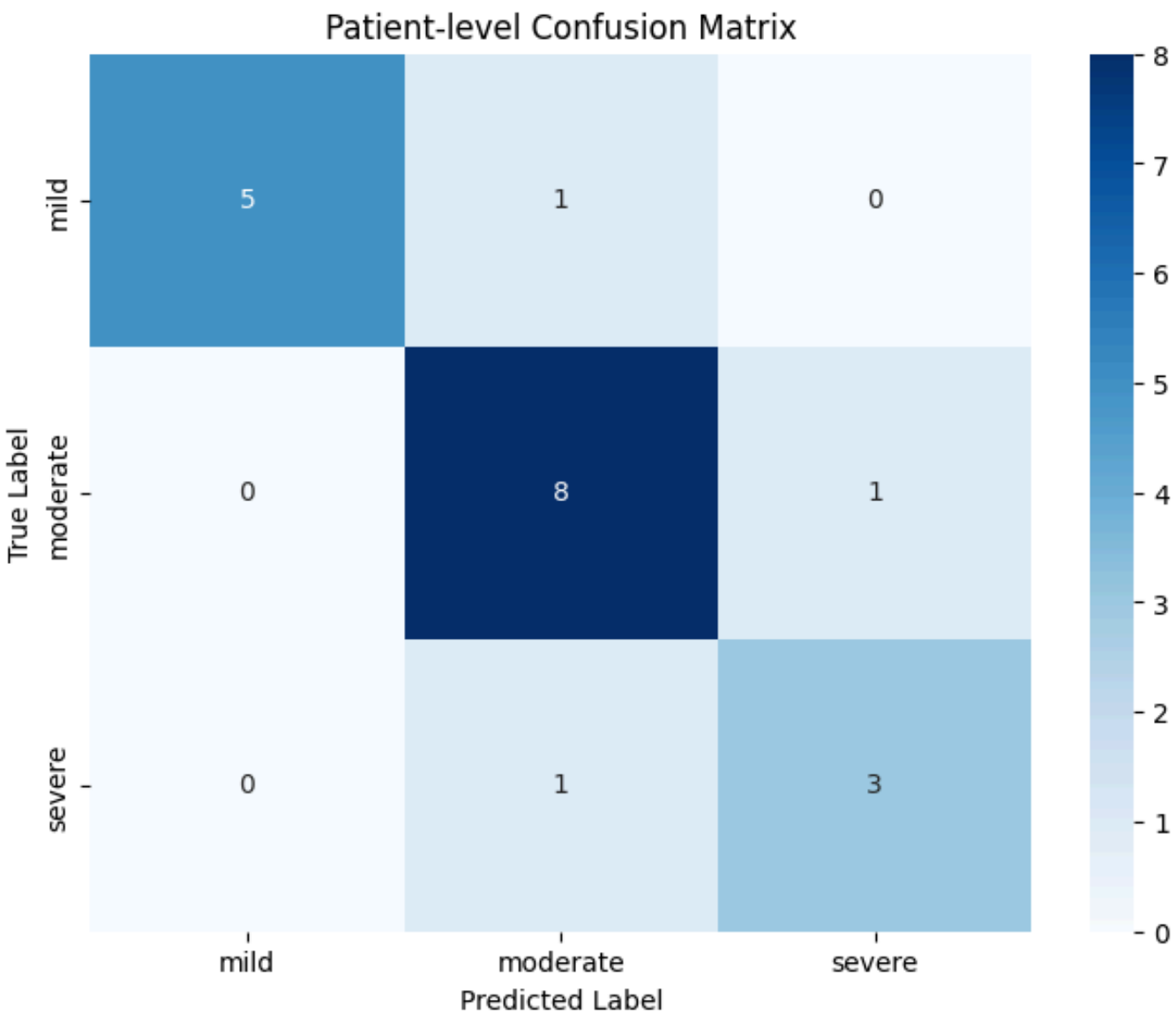
- Band Power: Delta, Theta, Alpha, Beta
- Both Absolute Power (AP) and Relative Power (RP)
- Extracted from selected EEG channels covering frontal, parietal, anterior, and posterior brain regions

Model:

- Random Forest classifier with optimised hyperparameters

Evaluation Results:

- Sample-level Accuracy: 78%
- Patient-level Accuracy: 84%



○

```
--- Sample-level Classification Report (Test Data) ---
              precision    recall  f1-score   support

   mild         0.77        0.85        0.81        114
  moderate         0.80        0.77        0.78        171
   severe         0.74        0.70        0.72         76

   accuracy              0.78        361
  macro avg         0.77        0.77        0.77        361
 weighted avg         0.78        0.78        0.78        361

Sample-level Accuracy: 0.778393351800554
```

```
--- Patient-level Classification Report ---
              precision    recall  f1-score   support

   mild         1.00        0.83        0.91         6
  moderate         0.80        0.89        0.84         9
   severe         0.75        0.75        0.75         4

   accuracy              0.84        19
  macro avg         0.85        0.82        0.83        19
 weighted avg         0.85        0.84        0.84        19

Patient-level Accuracy: 0.8421052631578947
```

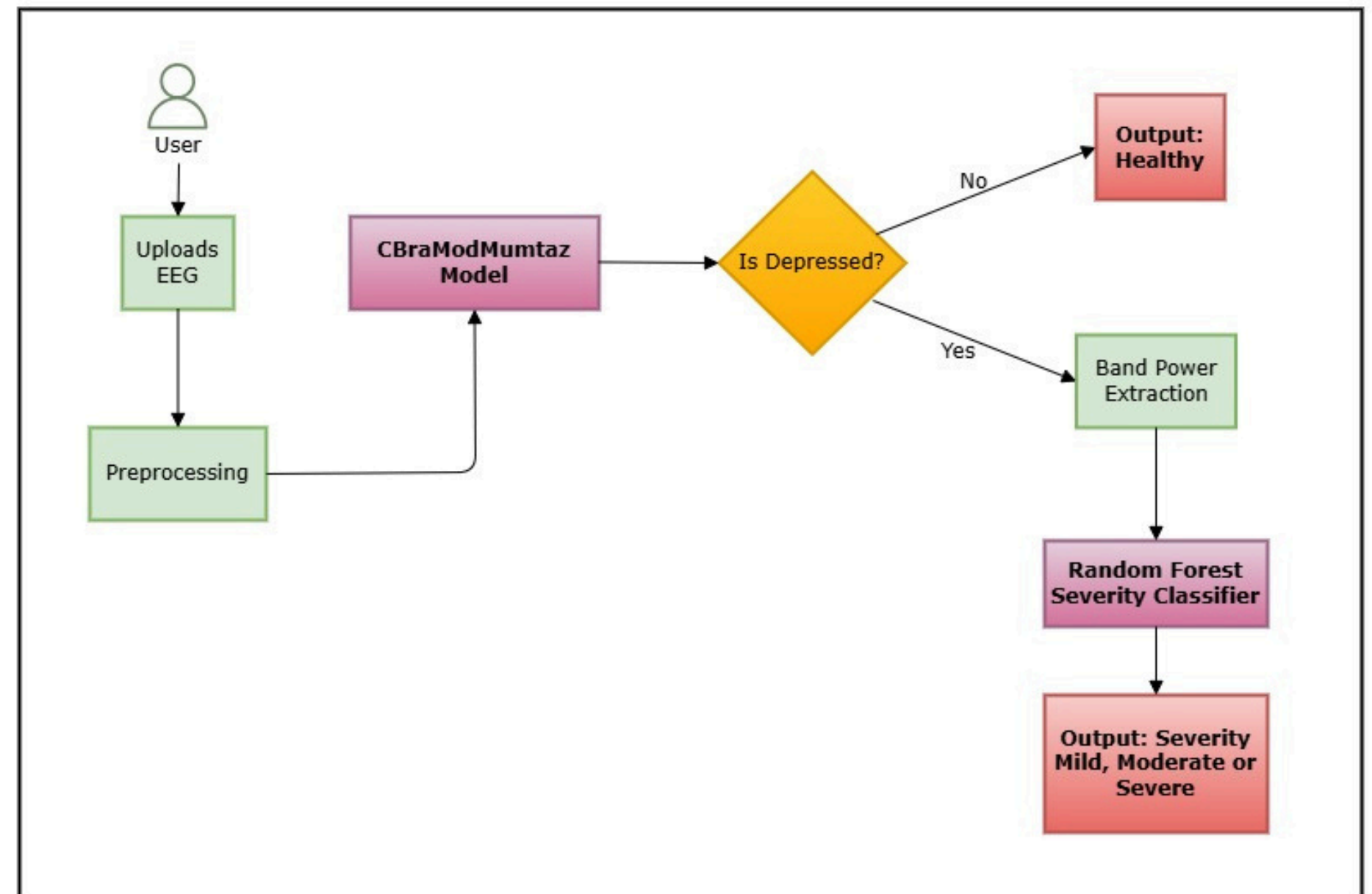
Module 2 - Depression Detection Module - 204110X

Final Hierarchical Classification Pipeline

Pipeline Blocks:

- Raw EEG → Preprocessing → CBraModMumtaz → Depression?
- If Depressed → Extract Band Power → XGBoost → Severity
 - CBraMod fine-tuned on MUMTAZZ
 - Severity classification trained on MODMA + BMC

- *Only EEGs classified as Depressed are passed to severity classifier.*
- *This avoids noisy label assumptions and improves generalization*



Module 2 - Depression Detection Module - 204110X

Related Works

- [1] Wu, C.-T., Huang, H.-C., Huang, S., Chen, I.-M., Liao, S.-C., Chen, C.-K., Lin, C., Lee, S.-H., Chen, M.-H., Tsai, C.-F., et al. (2021). Resting-State EEG Signal for Major Depressive Disorder Detection: A Systematic Validation on a Large and Diverse Dataset. *Biosensors*, 11(12), 499. <https://doi.org/10.3390/bios11120499>
- [2] Bader, M., Abdelwanis, M., Maalouf, M., & Jelinek, H. F. (2024). Detecting depression severity using weighted random forest and oxidative stress biomarkers. *Scientific Reports*, 14, Article 16328. <https://doi.org/10.1038/s41598-024-67251-y>
- [3] Khan, S., Umar Saeed, S. M., Frnda, J., Arsalan, A., Amin, R., Gantassi, R., & Noorani, S. H. (2024). A machine learning based depression screening framework using temporal domain features of the electroencephalography signals. *PLOS ONE*, 19(3), e0299127. <https://doi.org/10.1371/journal.pone.0299127>

Module 3 - Anxiety Detection Module - 204111C

Introduction

What is Anxiety?

- A mental health disorder characterized by excessive worry, fear, and nervousness.
- Affects daily life, work, and relationships.

Why Early Detection?

- Prevents severe mental health decline.
- Enables timely intervention and therapy.
- Reduces the risk of associated disorders like depression.

Objectives:

- Classify anxiety type using speech patterns, text sentiment, facial Expressions and physiological signals.
- Use Natural Language Processing (NLP) and Machine Learning (ML) for analysis.
- Support Neurologists with AI-driven assessments and personalized treatment suggestions



Module 3 - Anxiety Detection Module - 204111C

Research Gap

Although several systems exist for anxiety detection, few or none focus on classifying the type of anxiety.

No current system effectively combines Audio + Facial Expressions + Transcript Text in a multi-modal architecture for anxiety type detection.

Lack of labeled datasets and integrated multi-modal approaches limits the accuracy and clinical usefulness of existing solutions.

Module 3 - Anxiety Detection Module - 204111C

Dataset Overview: DAIC-WOZ

- Designed to support diagnosis of psychological distress, including Anxiety, Depression, and PTSD.
- Collected through Wizard-of-Oz clinical interviews using a virtual agent named Ellie, which was remotely operated by a human interviewer.
- The dataset includes 189 interview sessions, each ranging from 7 to 33 minutes (average: 16 minutes).
- Provides multi-modal data, including:
 - Audio recordings (speech features)
 - Video-based facial expressions (facial action units)
 - Interview transcripts (textual and linguistic features)
 - Questionnaire responses (psychological assessments)

Module 3 - Anxiety Detection Module - 204111C

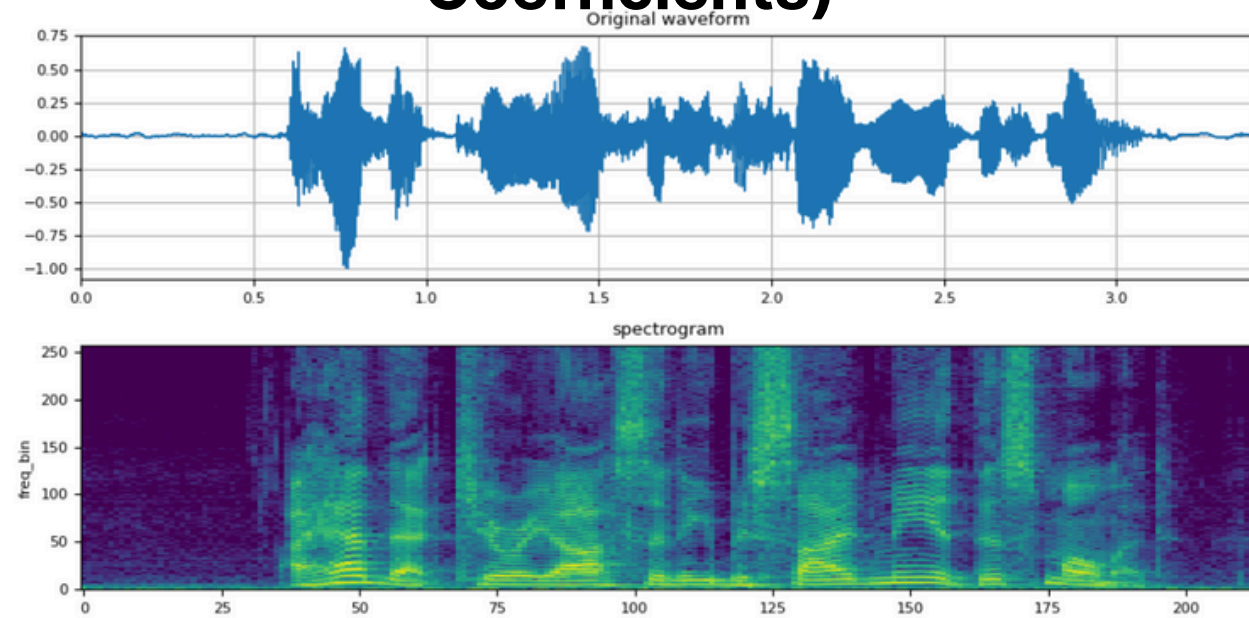
Input Features

Audio Features	Facial Features	Transcript Features
MFCCs (Mel Frequency Cepstral Coefficients) → Capture tone, pitch and stress patterns	Facial Action Units (AUs) via OpenFace → Micro-expressions (e.g., eyebrow raise, lip tightener)	Sentiment Polarity & Subjectivity → Negative or uncertain expressions indicate anxiety
Energy & Zero-Crossing Rate → Detect speaking intensity and nervous fluctuations	Blink Rate & Eye Gaze → Rapid blinking or gaze aversion are anxiety signs	Lexical Diversity → Repetitive or limited vocabulary shows cognitive strain
Speech Rate & Pauses → Slower or rushed speech can indicate anxiety	Head Movement Intensity → Restless or erratic movement may indicate discomfort	Part-of-Speech (POS) Tags → overuse of pronouns or modal verbs (“I think” , “maybe”)
Pitch Variability → High pitch variation often correlates with distress	Facial Tension → Tight lips, furrowed brows	Emotion Words → Frequency of fear, worry or stress related terms

Module 3 - Anxiety Detection Module - 204111C

System Workflow

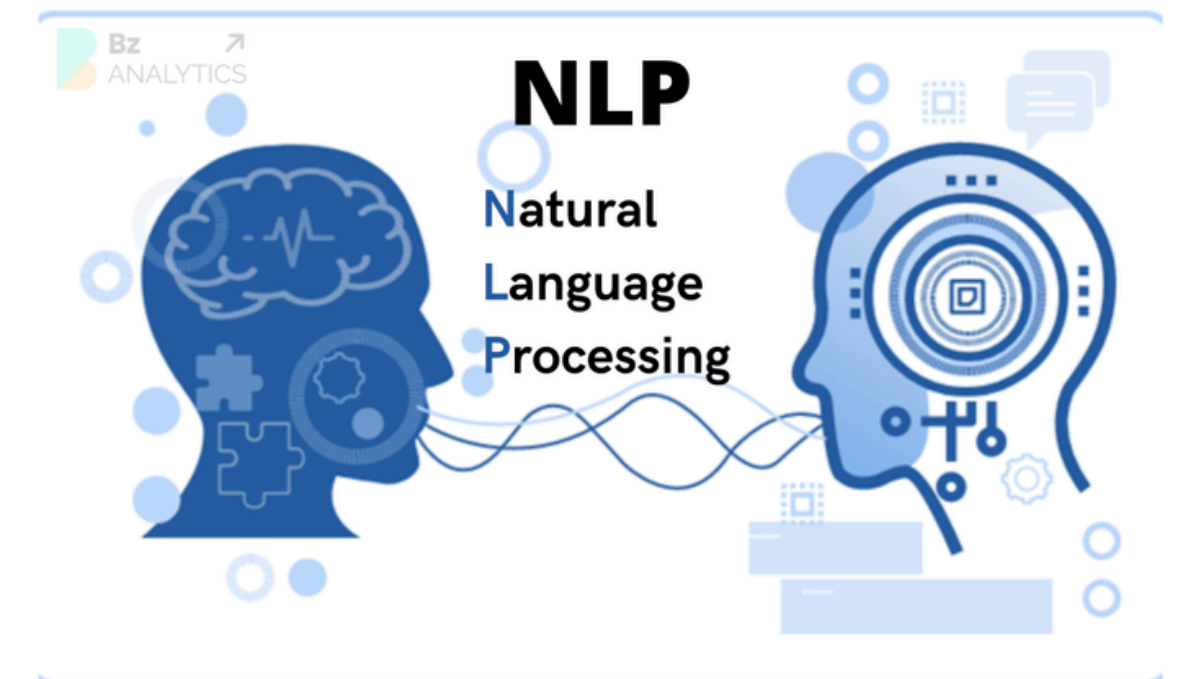
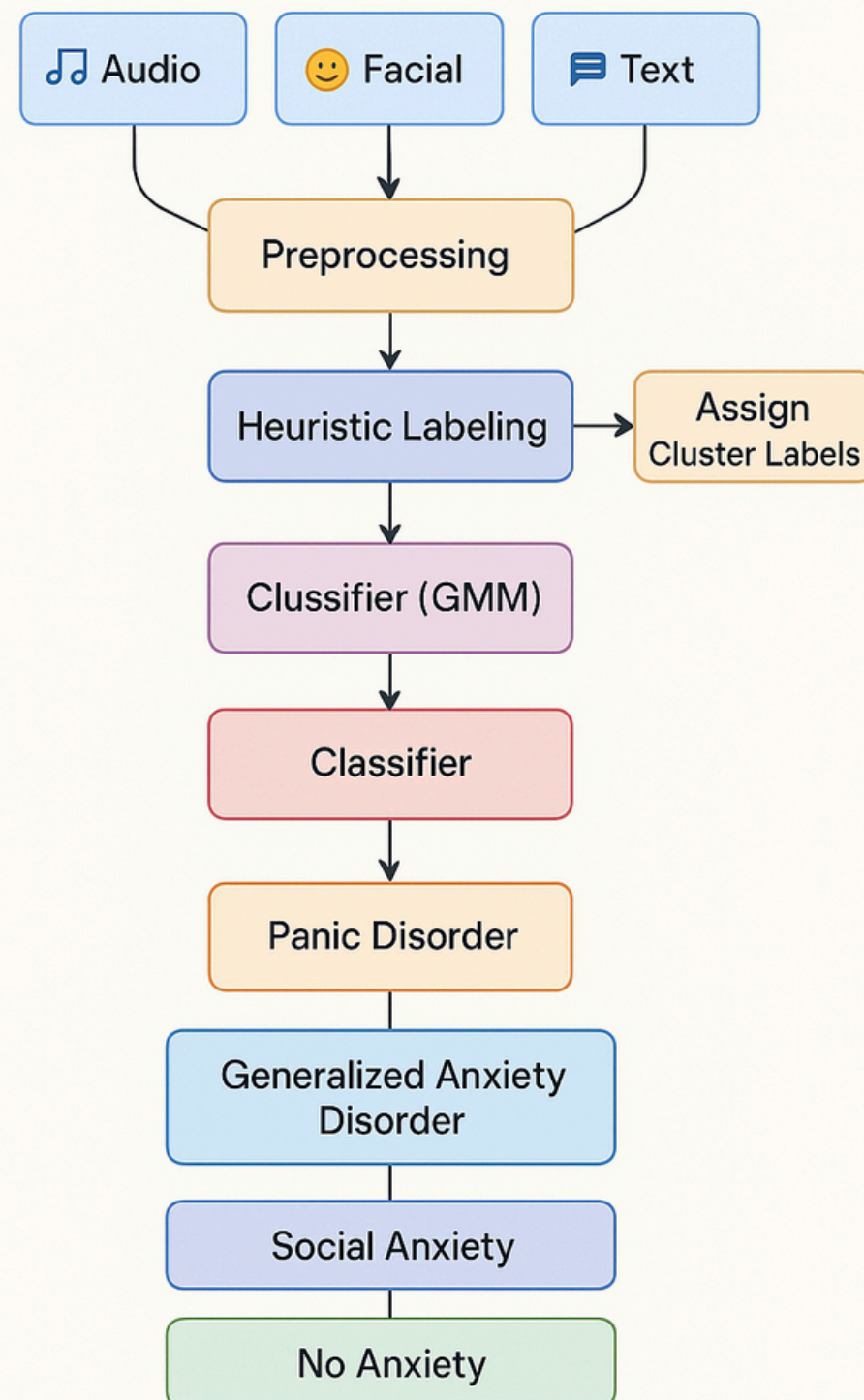
MFCCs (Mel-Frequency Cepstral Coefficients)



Facial Action Units



Heuristic + Clustering Anxiety Classification Pipeline



Module 3 - Anxiety Detection Module - 204111C

Process

Feature Extraction

Audio Features: .wav file

Facial Expression : CLNF_AUs, CLNF_Poses ,
CLNF_features

Transcript Features : .csv file

```
def extract_audio_features(audio_file_path):
    try:
        y, sr = librosa.load(audio_file_path, sr=None)
        mfcc_features = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=20)
        mfcc_means = mfcc_features.mean(axis=1)
        return pd.Series(mfcc_means)
    except Exception as e:
        print(f" Error processing {audio_file_path}: {e}")
        return None
```

```
def extract_facial_features_summary(participant_id):
    au_path = os.path.join(FACIAL_FEATURE_DIR, f"{participant_id}_CLNF_AUs.txt")
    features_path = os.path.join(FACIAL_FEATURE_DIR, f"{participant_id}_CLNF_features.txt")
    pose_path = os.path.join(FACIAL_FEATURE_DIR, f"{participant_id}_CLNF_pose.txt")

    au_df = load_AUs(au_path)
    features_df = load_facial_features(features_path)
    pose_df = load_pose(pose_path)

    if au_df is None or features_df is None or pose_df is None:
        return None

    try:
        merged_features_df = pd.concat([au_df, features_df, pose_df], axis=1)
        merged_features_df.replace([-100, np.inf, -np.inf], np.nan, inplace=True)
```

```
def extract_transcript_features(participant_id: str, transcript_text: str):
    doc = nlp(transcript_text)
    tokens = [token for token in doc if not token.is_punct and not token.is_stop and token.has_vector]

    word_count = len(tokens)
    avg_word_len = np.mean([len(token.text) for token in tokens]) if tokens else 0
    lexical_diversity = len(set([token.lemma_ for token in tokens])) / (word_count + 1e-6)

    pos_counts = doc.count_by(spacy.attrs.POS)
    pos_features = {f"pos_{doc.vocab[pos].text}": count for pos, count in pos_counts.items()}

    readability_score = textstat.flesch_kincaid_grade(transcript_text)

    sentiment = TextBlob(transcript_text).sentiment
    sentiment_polarity = sentiment.polarity
    sentiment_subjectivity = sentiment.subjectivity
```

Module 3 - Anxiety Detection Module - 204111C

Data Preprocessing & Cleaning

- Remove duplicate rows and invalid participant IDs
- Handle missing values with mean imputation
- Standardize all numeric features using StandardScaler
- Extract numeric IDs from participant identifiers

Feature Integration & Merging

- Inner join audio, facial, and transcript features on participant_id
- Remove temporal columns (start_time, stop_time)
- Create unified multimodal feature matrix

```
# === Use In-Memory Processed Feature Data ===
audio_df = audio_features_df          # audio processing pipeline
facial_df = final_facial_features_df  # facial feature extraction pipeline
transcript_df = df_transcript_summary # transcript processing pipeline

# === Clean IDs ===
def extract_numeric_id(val):
    if isinstance(val, str):
        match = re.search(r'\d+', val)
        return int(match.group()) if match else None
    return int(val)

audio_df["participant_id"] = audio_df["participant_id"].apply(extract_numeric_id)
transcript_df["participant_id"] = transcript_df["participant_id"].apply(extract_numeric_id)
facial_df["participant_id"] = facial_df["participant_id"].apply(extract_numeric_id)

# === Merge All Features ===
df = audio_df.merge(transcript_df, on="participant_id", how="inner")
df = df.merge(facial_df, on="participant_id", how="inner")
df = df.drop(columns=['participant_id', 'start_time', 'stop_time'], errors='ignore')
```

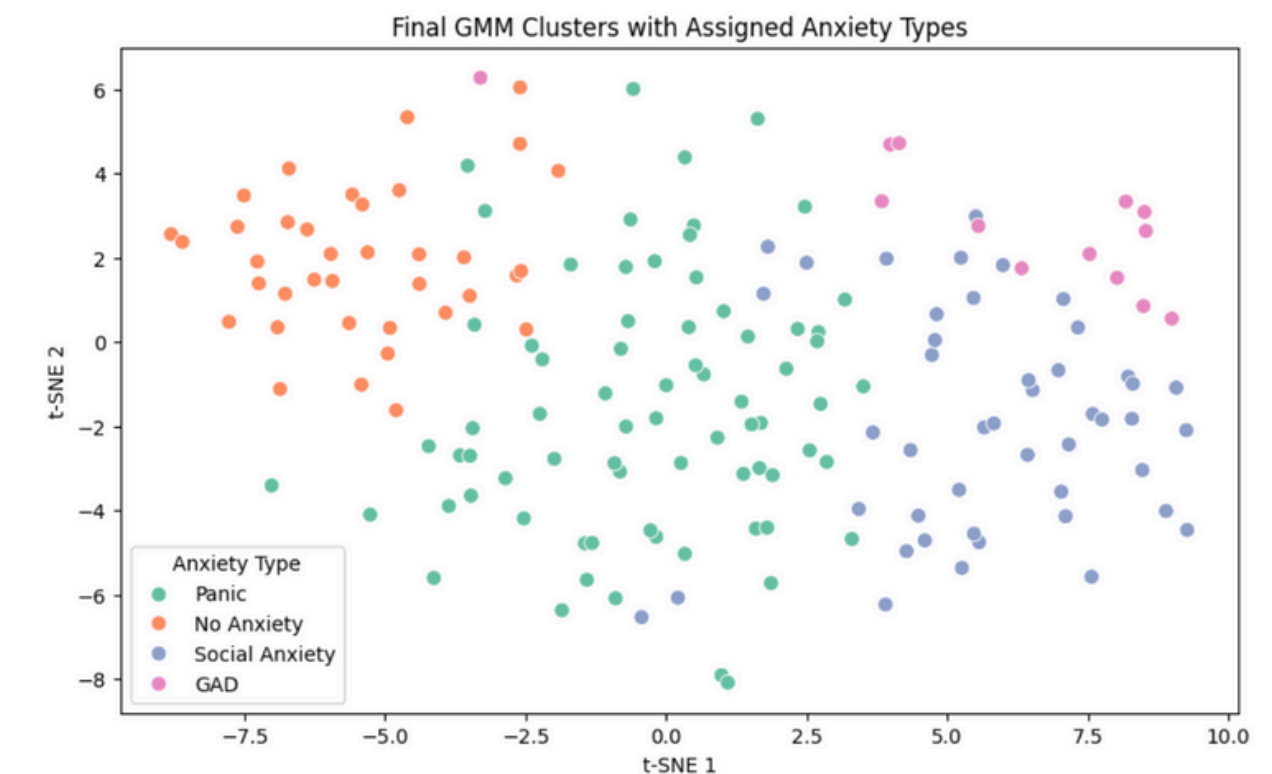
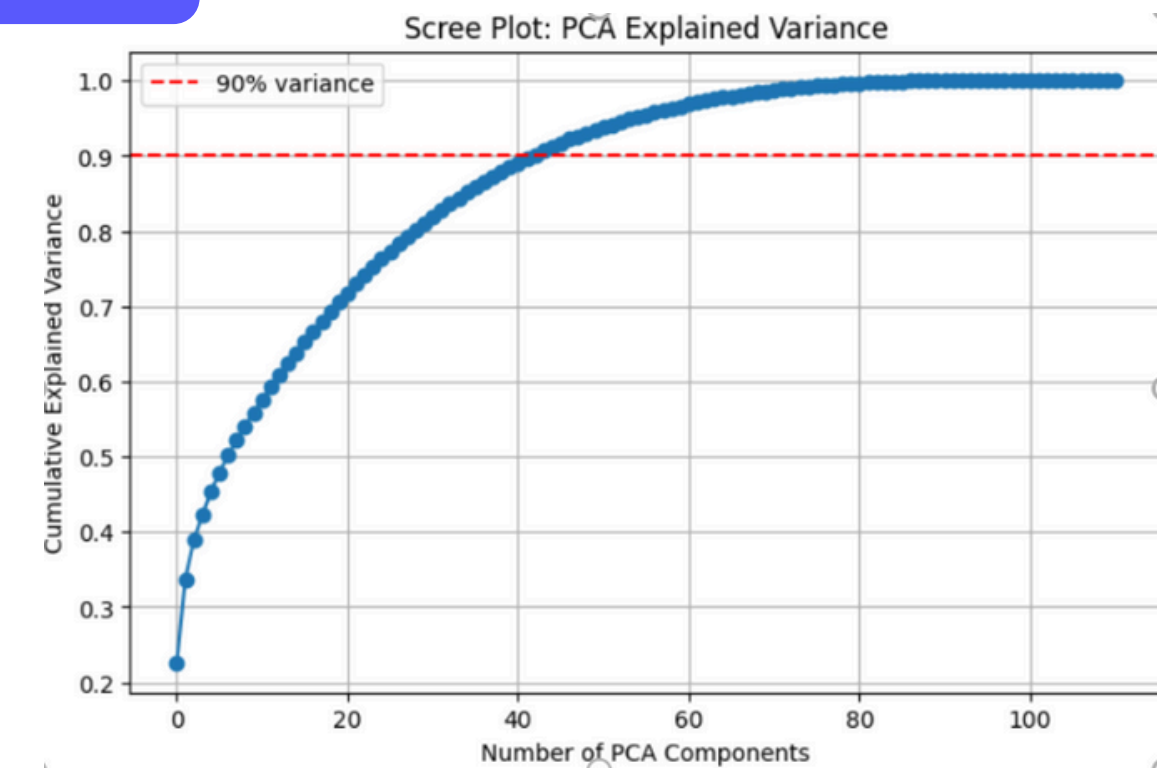
Module 3 - Anxiety Detection Module - 204111C

Dimensionality Reduction & Visualization

- Apply PCA to retain 95% of variance
- Use t-SNE for 2D visualization
- Reduce computational complexity
- Preserve important feature relationships

Unsupervised Clustering

- Apply GMM with 4 components for anxiety types
- Generate cluster assignments for each participant
- Analyze cluster characteristics and distinctive features
- Visualize clusters using t-SNE scatter plot



Module 3 - Anxiety Detection Module - 204111C

Ensemble Classification Model

Random Forest

n_estimators=100

class_weight='balanced'

XGBoost

eval_metric='mlogloss'

use_label_encoder=False

MLP Neural Network

hidden_layers= (64, 32)

max_iter=500

```
# Define base models
rf = RandomForestClassifier(n_estimators=100, random_state=42, class_weight='balanced')
xgb = XGBClassifier(eval_metric='mlogloss', use_label_encoder=False, random_state=42)
mlp = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state=42)

# Define ensemble
ensemble = VotingClassifier(estimators=[
    ('rf', rf),
    ('xgb', xgb),
    ('mlp', mlp)
], voting='soft')

# Train ensemble
ensemble.fit(X_train_scaled, y_train)

# Predict
y_pred = ensemble.predict(X_test_scaled)
```

Module 3 - Anxiety Detection Module - 204111C

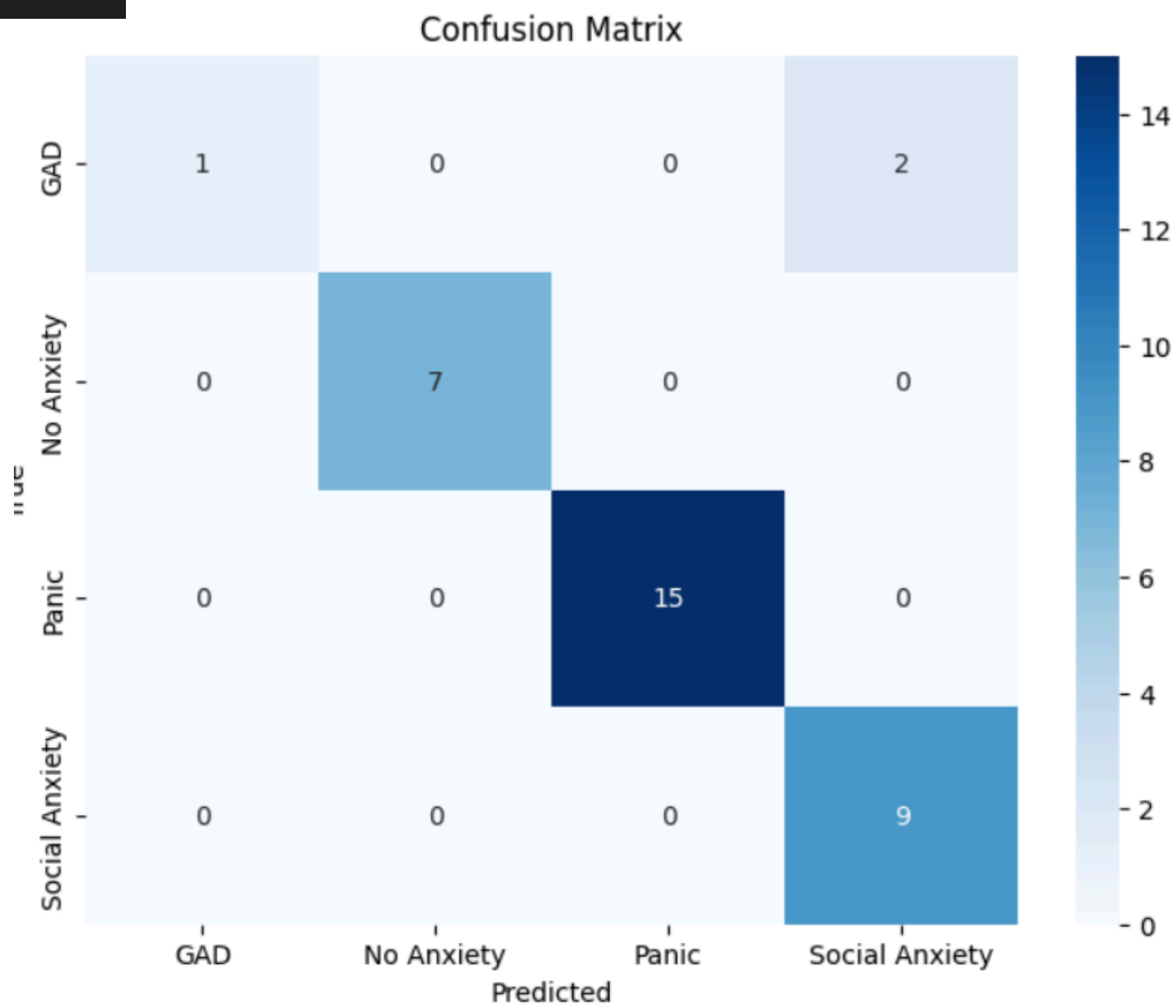
Evaluation

```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded, test_size=0.2, random_state=42, stratify=y_encoded
)
```

✓ Ensemble Classification Report:

	precision	recall	f1-score	support
General Anxiety Disorder	0.88	0.93	0.90	15
No Anxiety	0.89	0.89	0.89	9
Panic Disorder	1.00	1.00	1.00	7
Social Anxiety	1.00	0.67	0.80	3
accuracy			0.91	34
macro avg	0.94	0.87	0.90	34
weighted avg	0.92	0.91	0.91	34

✓ Ensemble Accuracy: 0.9118



Module 3 - Anxiety Detection Module - 204111C

Anxiety Classification Tool

Upload Audio (.wav):

368_AUDIO.wav

Facial Expression Text Files:

368_CLNF_AUs.txt

368_CLNF_features.txt

368_CLNF_pose.txt

Transcript File (.csv):

368_TRANSCRIPT.csv

Prediction Result

☒ **Anxiety Type: Generalized Anxiety Disorder**

Module 3 - Anxiety Detection Module - 204111C

Novelty of the Research Module

Learning Approach :- Hybrid (Clustering + Classification)

Data Fusion :- Multimodal: Audio + Facial + Transcript

Interpretability :- Z-score-based statistical labeling

Clinical Alignment :- Psychology-informed anxiety type mapping

Model Design :- Ensemble Voting Classifier with robust learners

Inclusiveness :- Handles underrepresented anxiety subtypes

Module 3 - Anxiety Detection Module - 204111C

Related Works

[1]

P. Divyashree, G. Yadav A, N. Jayadev, and S. Chidaravalli, "Stress and Anxiety Detection through Speech Recognition Using Deep Neural Network." Accessed: Nov. 14, 2024. [Online]. Available: https://ijirt.org/publishedpaper/IJIRT154638_PAPER.pdf

[2]

H. Mo, Siu Cheung Hui, X. Liao, Y. Li, W. Zhang, and S. Ding, "A Multimodal Data-Driven Framework for Anxiety Screening," IEEE transactions on instrumentation and measurement, vol. 73, pp. 1–13, Jan. 2024, doi: <https://doi.org/10.1109/tim.2024.3352713>.

[3]

M. Gavrilescu and N. Vizireanu, "Predicting Depression, Anxiety, and Stress Levels from Videos Using the Facial Action Coding System," Sensors, vol. 19, no. 17, p. 3693, Aug. 2019, doi: <https://doi.org/10.3390/s19173693>.

[4]

Y. Chen, Y. Tang, Q. Ni, Y. Gao, and C. Wang, "Integrated multimodal analysis for high-accuracy anxiety disease subtype classification," Psychiatry Research: Neuroimaging, vol. 349, p. 111991, Apr. 2025, doi: <https://doi.org/10.1016/j.psychresns.2025.111991>.

Further Work



Temporal Modelling:

- Use LSTM/GRU to capture changes in anxiety over time from facial, audio, and text inputs.

Real-Time Detection:

- Deploy lightweight models (e.g., ONNX, TFLite) for mobile or web-based real-time mental health tools.

Physiological Signals:

- Add data like heart rate or EEG from wearables to improve accuracy and detect internal anxiety cues.

Dataset Expansion & Standardization:

- Build or contribute to a larger, multi-center EEG dataset with well-annotated depression severity labels (e.g., PHQ-9, HAM-D).

Challenges

1. Technical Limitations

- Limited, biased, and small psychiatric datasets.
- Difficulty in integrating EEG, MRI, speech, and behavioral data.
- AI models lack transparency (black-box issue).

2. Ethical & Privacy Limitations

- AI bias can lead to misdiagnosis in diverse populations.
- Unclear legal responsibility for AI errors.
- Sensitive mental health data requires strict security.

3. Medical & Clinical Challenges

- Hard to distinguish psychiatric from neurological disorders.
- Symptoms vary across individuals and cultures.
- No universal psychiatric diagnosis criteria.

Conclusion

- **Enhanced Diagnostic Accuracy** - AI leverages ML and NLP to analyze patient interactions, improving psychiatric disorder detection.
- **Real-Time Therapeutic Support** - AI can monitor emotional well-being and provide personalized interventions like cognitive exercises.
- **Sentiment Analysis for Mental Health** - AI detects emotional states (e.g., sadness, anger) from text data, enabling tailored therapy.
- **Challenge of Neurological-Psychiatric Overlap** - Many neurological disorders (e.g., Parkinson's, epilepsy) co-occur with psychiatric conditions, complicating diagnosis.
- **Need for Integrated AI Models** – Combining neurological and psychiatric data can improve diagnosis, treatment planning, and holistic patient care.

ICTIS
THAILAND 2025



CERTIFICATE

This is to certify that

**H. K. Buddhima Kaushalya, N. G. W. Gamage, D.K.O.P. Kumarathunga,
K. A. Dilini T. Kulawansa, S. C. Premaratne**

has contributed a paper titled

AI Driven Support System for Neurologists Integrating Neurological and Psychiatric Data

in the

**9th International Conference on
Information and Communication Technology for Intelligent Systems**

The paper has also been selected for publication in the (ICTIS) Conference as per fulfilment of guidelines issued by Springer.

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TPC CHAIR

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TACHANUN KANGWANTRAKOOL, Ph.D
LOCAL CONFERENCE CHAIR

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AMIT JOSHI, Ph.D
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References

- [1] S. Tutun et al., "An AI-based Decision Support System for Predicting Mental Health Disorders," Information Systems Frontiers, vol. 25, no. 3, May 2022, doi: <https://doi.org/10.1007/s10796-022-10282-5>.
- [2] Md. M. Islam, S. Hassan, S. Akter, F. A. Jibon, and Md. Sahidullah, "A comprehensive review of predictive analytics models for mental illness using machine learning algorithms," Healthcare Analytics, p. 100350, Jun. 2024, doi: <https://doi.org/10.1016/j.health.2024.100350>. Disorders,"
- [3] M. Tanaka and L. Vécsei, "From Lab to Life: Exploring Cutting-Edge Models for Neurological and Psychiatric Disorders," Biomedicines, vol. 12, no. 3, p. 613, Mar. 2024, doi: <https://doi.org/10.3390/biomedicines12030613>.
- [4] A. Rizea, "Deep Learning-based Solution for Mental Health Issues," Database Systems Journal, vol. XIII, no. 57, 2022, pp. 57–59, doi: [https://www.dbjournal.ro/archive/33/33_7.pdf].
- [5] U. Rana and R. Singh, "The Role of Artificial Intelligence in Mental HealthCare," Dr. Harisingh Gour Vishwavidyalaya (A Central University), Sagar, India, 2023, doi: [<https://osf.io/preprints/socarxiv/r4umy>].
- [6] M. Terra, M. Baklola, S. Ali, and K. El-Bastawisy, "Opportunities, applications, challenges and ethical implications of artificial intelligence in psychiatry: a narrative review," Egyptian Journal of Neurology, Psychiatry and Neurosurgery, vol. 59, no. 80, 2023, doi:<https://ejnnpn.springeropen.com/articles/10.1186/s41983-023-00681-z>

References

- [7] Hasan Alkahtani, Theyazn H. H. Aldhyani, and Ahmed Abdullah Alqarni, "Artificial Intelligence Models to Predict Disability for Mental Health Disorders," Deleted Journal, vol. 3, no. 3, Jan. 2024, doi: <https://doi.org/10.57197/jdr-2024-0022>.
- [8] G. Parimala, R. Kayalvizhi, and S. Nithiya, "Mental Health: Detection & Diagnosis," 2022 International Conference on Computer Communication and Informatics (ICCCI), pp. 1–6, Jan. 2022, doi: <https://doi.org/10.1109/iccci54379.2022.9740834>.
- [9] C. Xue et al., "AI-based differential diagnosis of dementia etiologies on multimodal data," Nature Medicine, pp. 1–13, Jul. 2024, doi: <https://doi.org/10.1038/s41591-024-03118-z>.
- [10] Krishnaraj Chadaga et al., "SADXAi: Predicting social anxiety disorder using multiple interpretable artificial intelligence techniques," SLAS TECHNOLOGY, vol. 29, no. 2, pp. 100129–100129, Apr. 2024, doi: <https://doi.org/10.1016/j.slant.2024.100129>.
- [11] A. Javeed, A. L. Dallora, J. S. Berglund, A. Ali, L. Ali, and P. Anderberg, "Machine Learning for Dementia Prediction: A Systematic Review and Future Research Directions," Journal of Medical Systems, vol. 47, no. 1, Feb. 2023, doi: <https://doi.org/10.1007/s10916-023-01906-7>.
- [12] Shusharina, N.; Yuxhnenko, D.; Botman, S.; Sapunov, V.; Savinov, V.; Kamyshov, G.; Sayapin, D.; Voznyuk, I. Modern Methods of Diagnostics and Treatment of Neurodegenerative Diseases and Depression. Diagnostics 2023, 13, 573. <https://doi.org/10.3390/diagnostics13030573>

References

- [13] Termine, A., Fabrizio, C., Strafella, C., Caputo, V., Petrosini, L., Caltagirone, C., Giardina, E., & Cascella, R. (2021). Multi-Layer Picture of Neurodegenerative Diseases: Lessons from the Use of Big Data through Artificial Intelligence. *Journal of Personalized Medicine*, 11(4), 280. <https://doi.org/10.3390/jpm11040280>
- [14] Wysa, "Wysa - Everyday mental health," Wysa - Everyday Mental Health, 2023. <https://www.wysa.com/>
- [15] Youper, "Youper - Emotional Health Assistant Powered by AI," Youper, 2019. <https://www.youper.ai/>
- [16] "DxGPT: Diagnostic decision support software based on GPT-4," DxGPT: Diagnostic decision support software based on GPT-4, 2025. https://dxgpt.app/?gad_source=1&gclid=CjwKCAjwp8--BhBREiwAj7og149UKhApVZPYh_dfN06LRiFOAnSS9boknZMpRytaIX_wagYFu3cTkRoCGwwQAvD_BwE (accessed Mar. 14, 2025).



Team Mavericks

Thank You